

Arnav Samal

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EDUCATION

Carnegie Mellon University - School of Computer Science

Master of Science in Computer Vision (Robotics Institute)

Relevant Coursework: Advanced Computer Vision, Visual Learning & Recognition, Learning for 3D Vision

Pittsburgh, PA

Dec 2027

National Institute of Technology, Rourkela

Bachelor of Technology, Computer Science & Engineering | GPA: 9.13/10.0 | Dept Rank: 7/153

Relevant Coursework: Computer Vision, Deep Learning, Machine Learning, Natural Language Processing

Rourkela, India

May 2026

SKILLS

Programming Languages: Python, C++, C, SQL

Tools: Linux, Git, Conda, Weights & Biases, Docker, Jupyter, LaTeX

Libraries/Frameworks: PyTorch, OpenCV, HuggingFace, Sci-kit Learn, SciPy, NumPy, Matplotlib, Flask

Computer Vision & ML: Vision-Language Models (VLMs), Diffusion Models, Vision Transformers, Self-Supervised Learning, Explainability (CBMs, Post-hoc XAI)

RESEARCH EXPERIENCE

IISc Bangalore | Visual Computing Lab, Dept. of CDS

Research Intern (Advisor: Dr. Anirban Chakraborty)

Bengaluru, India

Jan 2026 – Jun 2026

- Engineered a distributed Federated Learning pipeline to isolate how overparameterization (up to ResNet-101) mitigates client drift, benchmarking 4 SOTA algorithms (FedAvg, FedDyn, SCAFFOLD, FedProx) across CIFAR-10/100 and TinyImageNet in highly non-IID settings.
- Developed SurfScaff3D, a training-free text- and image-to-3D framework built on hierarchical B-spline surface patches for editable, open-domain 3D object generation.

IIT Hyderabad | Machine Learning & Vision Group, Dept. of CSE

Research Intern (Advisor: Prof. Vineeth N Balasubramanian, with Microsoft Research)

Hyderabad, India

May 2025 – Dec 2025

- Investigated concept quality in CBMs by analyzing concept locality to diagnose concept leakage; processing 2TB+ medical imaging data and applying locality-to-concept specific metrics.
- Found that CBM models attend 45% more to spurious background regions than diagnostically relevant features, exposing a key medical-imaging failure mode.

IIT Hyderabad | Data-Driven Intelligence & Learning Lab, Dept. of AI

Research Intern (Advisor: Dr. Konda Reddy Mopuri)

Hyderabad, India

May 2024 – Aug 2024

- Benchmarked post-hoc explainability techniques (attention rollout, token pruning) across Vision Transformer architectures to quantify interpretability-efficiency tradeoffs.
- Implemented token pruning strategies using top-K token overlay and patch importance metrics, reducing inference compute by 66% while keeping classification accuracy within 1-3% of the baseline.

PUBLICATIONS

- Anay Loya*, Rishi Gupta*, **Arnav Samal**, and Anirban Chakraborty. "SurfScaff3D: Bridging Geometric and Semantic Representations for LLM-based 3D Object Generation." *WACV 2027 (Under Review)*

PROJECTS

Explainable Sketch to Face Generation – NIT Rourkela, Research Project | Code

Aug 2025 – Nov 2025

- Fine-tuned Stable Diffusion v1.5 using ControlNet and LoRA, achieving an FSIM of 0.7386 to outperform baseline architectures in structural fidelity.
- Engineered a temporal XAI framework adapting Grad-CAM to regression U-Nets, mapping gradient attribution across DDIM sampling timesteps to visualize feature-anchored sketch-to-face reasoning.

Capsule Vision Challenge 2024 – Global Competition | Preprint, Code

Aug 2024 – Oct 2024

- Architected a multi-model CNN and Vision Transformer ensemble pipeline, mitigating severe class imbalance via focal loss optimization and weighted sampling strategies.
- Secured global Rank 5 out of 500+ participating teams.